Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/18/2006
PI Name:	Shaqfeh, Eric S.G. Ph.D.		
Project Title:	Drop Breakup and DNA Advection in Fixed Beds of Fibers: Hairpin Dynamics in Ordered and Disordered Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	esgs@stanford.edu	**Fax:**	FY (650) 723-9780
PI Organization Type:	UNIVERSITY	**Phone:**	(650) 723-3764
Organization Name:	Stanford University		
PI Address 1:	Stauffer III, 207		
PI Address 2:			
PI Web Page:			
City:	Stanford	**State:**	CA
Zip Code:	94305-5025	**Congressional District:**	14
Comments:	Email changed 2/2008 from eric@chemeng.stanford.edu		
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	03/07/2002	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Singh, Bhim S	**Contact Phone:**	216-433-5396
Contact Email:	Bhim.S.Singh@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2724		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	"Hairpin dynamics" is a hall mark of many separation devices including new gene sequencing and sieving devices that have application to space-based biosensors. This project focusses on understanding these dynamics in the flows of drops and polymer molecules with application to creating these sennsors. We are particularly working on sequence specific DNA sensors for use in space to sense DNA damage by HZE particles. To gain a fundamental understanding of Drop and DNA dynamics in the flow through obstacle arrays (including fiber obstacles) focussing on how "hairpin dynamics" creates drop breakup and polymer mobility changes. This information is used directly then in optimizing two new DNA sequence sensore designs. The significance for the project is really split into to parts. For drop dynamics in the flow through obstacles, the task is a fundamental study of drop breakup and emulsification through fiber arrays with the goal of controlling drop-size distributions. For DNA dynamics, the task is focussed on a understanding and engineering obstacle arrays for sieving, stretching and sequencing purposes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In response to the need for research focused on life-support for the upcoming “Mission to Mars”, we have now directed our efforts toward developing early warning systems for detection of astronaut DNA damage by HZE particle exposure. It is well known and described clearly in the “Critical Road Map” that the biological risks of continuous exposure to high energy galactic cosmic ray particles include DNA double strand breaks and deletsions. It is generally agreed that these breaks are early events leading to malignant transformations and carcinogenesis. We have focused on developing two devices which are designed for sequence specific single DNA molecule sensing. Both of these devices rely on stretching dsDNA and binding (via sequence specific oligomer linkers) fluorescent beads to the molecules in a continuous flow sequence.
Bibliography Type:	Description: (Last Updated: 02/27/2004)